



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 1

2019 Year 11 Course Assessment Task 2

Wednesday, 29 May 2019

General instructions

- Working time – 1 hour (plus 5 minutes reading time).
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- Attempt all questions.
- All necessary working should be shown in every question.
- Marks may be deducted for illegible or incomplete working.

NESA STUDENT #: # BOOKLETS USED:

Class: (please ✓)

☐ 11MAT.1 – Mr Lam

☐ 11MAT.2 – Mr Sekaran

☐ 11MAT.3 – Ms Park

☐ 11MAT.4 – Ms Ham

☐ 11MAT.5 – Ms Gan

Marker's use only

QUESTION	1	2	3	4	Total	%
MARKS	$\frac{22}{22}$	$\frac{11}{11}$	$\frac{14}{14}$	$\frac{13}{13}$	$\frac{60}{60}$	

Question 1 (22 marks)**Marks**

- (a) Is the function
- $f(x) = x^3 - x$
- odd, even or neither? Justify your answer.

2

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- (b) Find the domain and range of the following in interval notation.

i. $y = 3x^2 - 1$

2

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ii. $y = \frac{1}{x-2}$

2

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iii. $y = -\sqrt{9 - x^2}$

2

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- (c) Given
- $f(x) = x^2 + 1$
- and
- $g(x) = 3 - x$
- , find:

i. $f(g(2))$.

1

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ii. $g(f(x))$ in simplest form.

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iii. The value(s) of x such that $g(f(x)) = f(x)$.

2

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(d) Solve $|7 - 3x| = 1$

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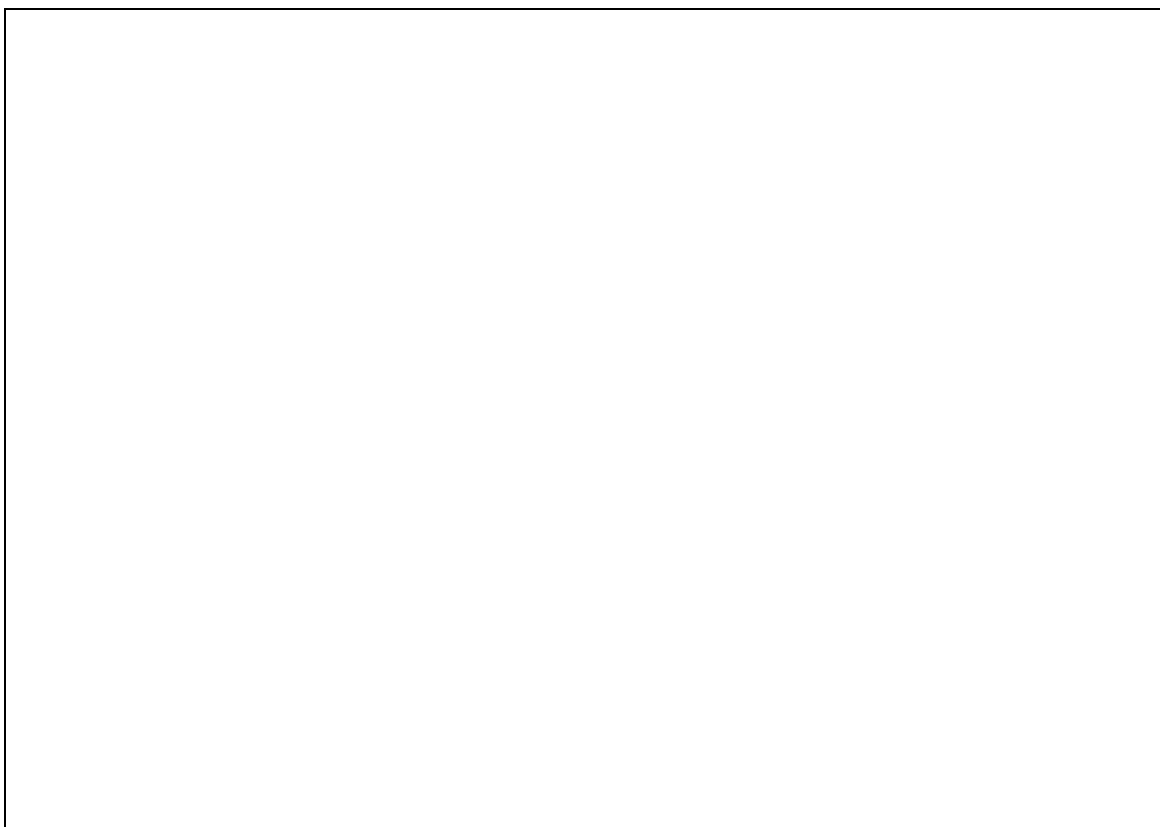
(e) Sketch the following graphs showing all important features, including x and y intercepts.

i. $y = (x - 1)(x + 3)(4 - x)$

2

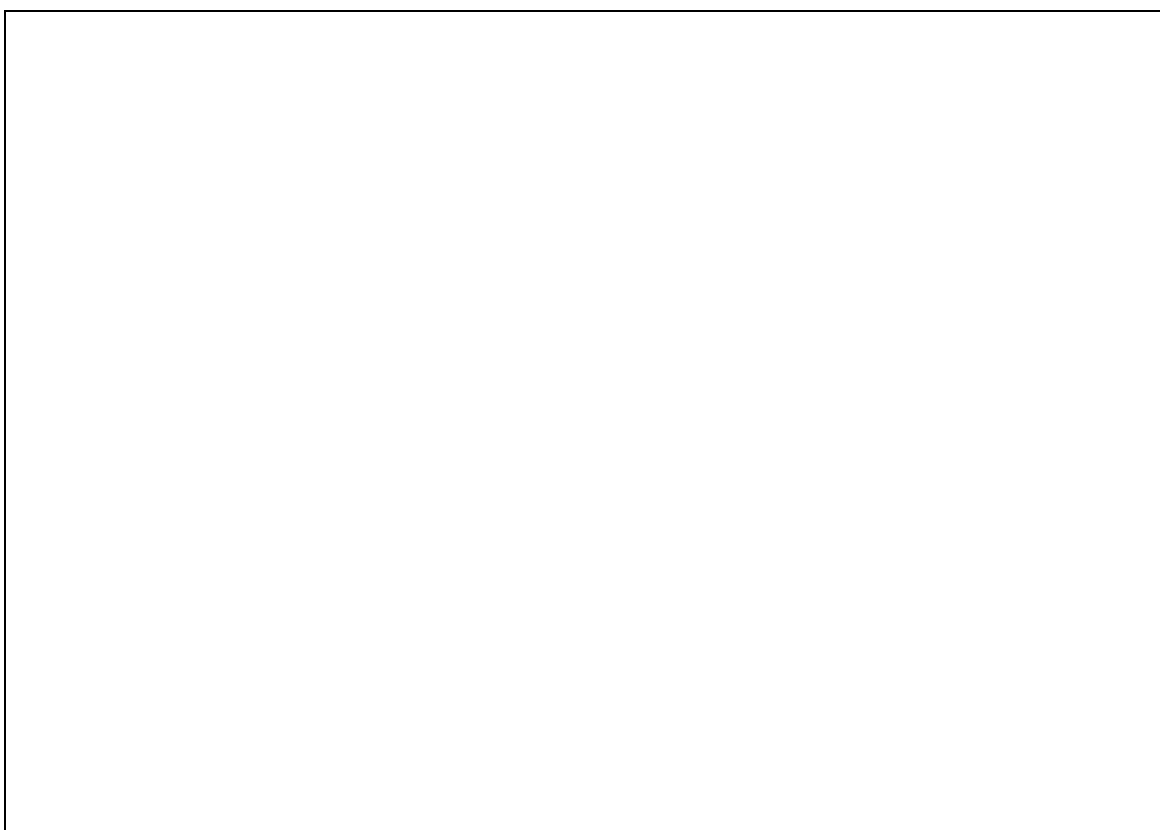
ii. $y = -x^2 - 4x + 5$

2



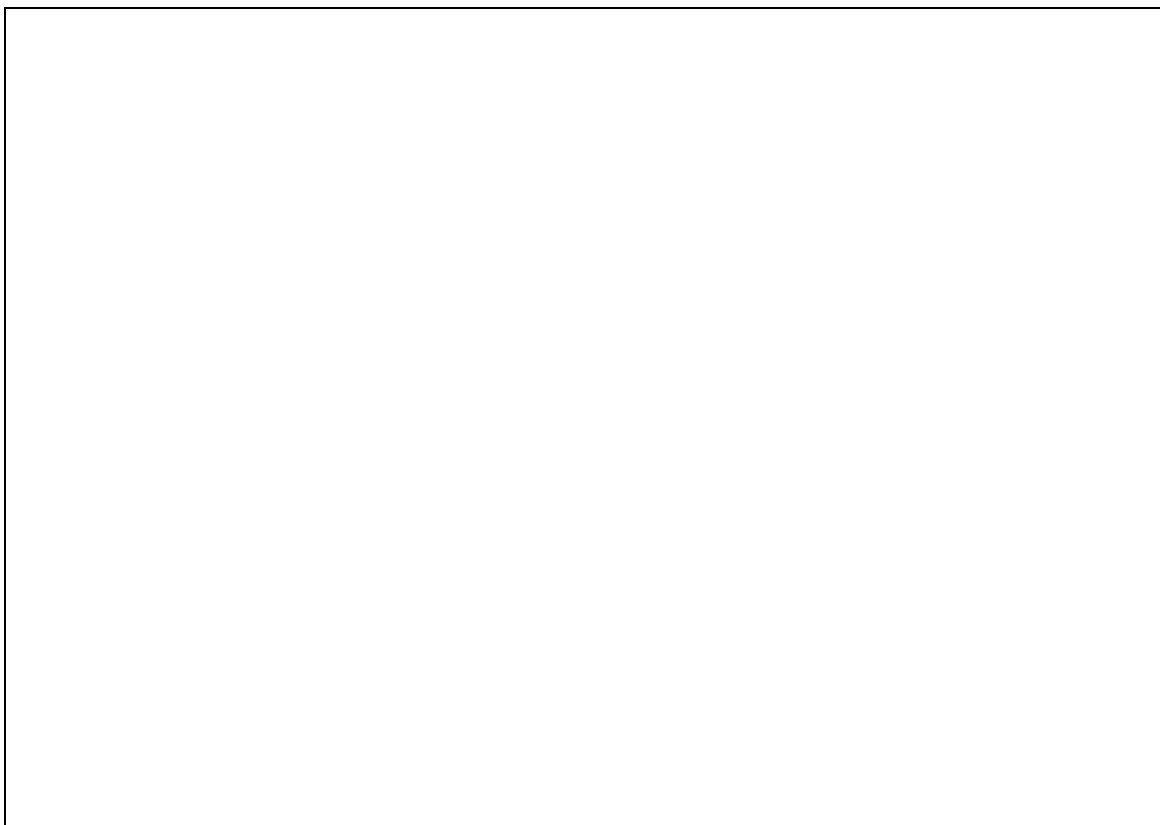
iii. $y = \sqrt{x + 3}$

2



iv. $y = |x - 2| + 1$

2



Examination continues overleaf

Question 2 (11 marks)**Marks**

- (a) Solve the equation
- $\log_2 x + \log_2(x + 7) = 3$
- .

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- (b) Fully simplify
- $\sqrt{\frac{3^{5k+2}}{27^k}}$

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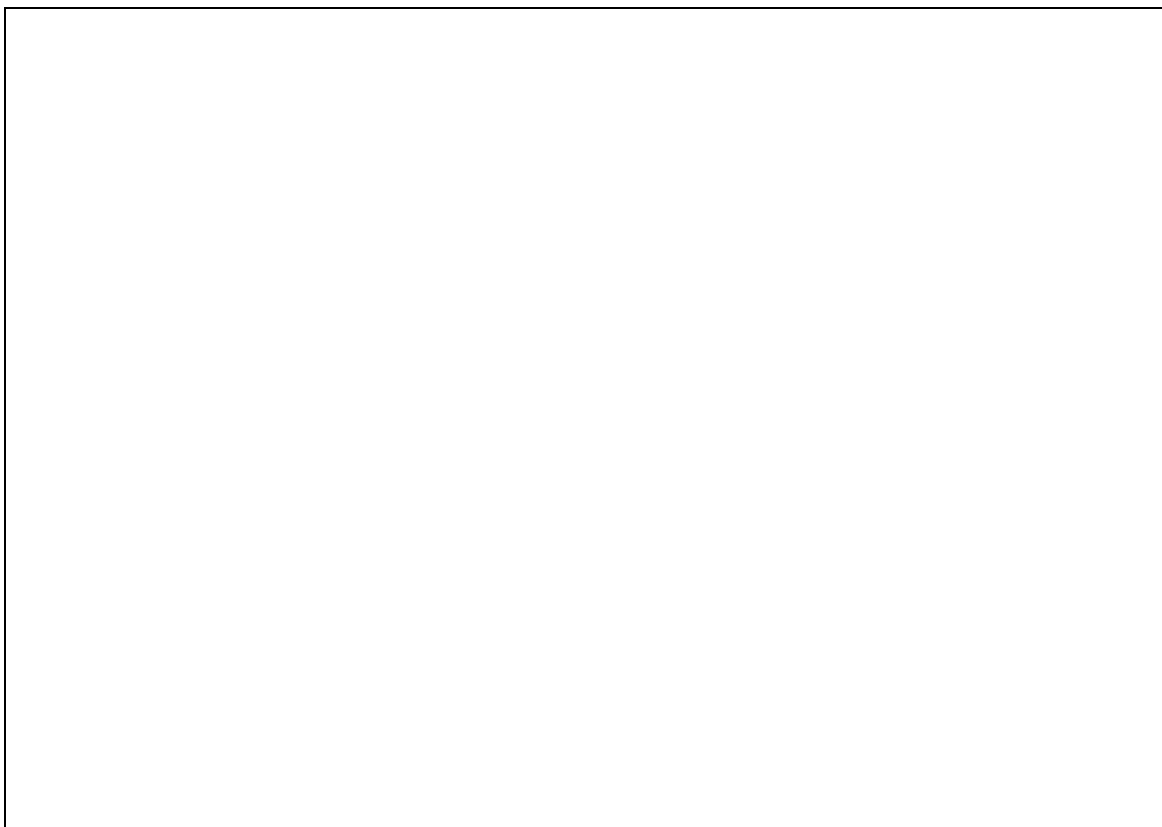
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- (c) Sketch the graph of $y = \log_4(3 - x)$, showing all important features.

2



- (d) The number N of individuals in a population at time t years is given by

$$N = 100 - 60e^{-0.1t} \quad \text{where } e = 2.71828 \dots$$

- i. Find the initial population N at $t = 0$ and what N tends towards as $t \rightarrow \infty$.

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- ii. Find the exact time taken for the population to double its initial size.

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Question 3 (14 marks)**Marks**

- (a) If $\sin \theta = \frac{1}{2}$ and $\cos \theta < 0$, what is the exact value of $\tan \theta$?

2

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- (b) Solve $\cos^2 x + 2\sin x = 0$, for $0^\circ \leq x \leq 360^\circ$.

2

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- (c) Solve $\tan 2x = -\sqrt{3}$, for $-180^\circ \leq x \leq 180^\circ$.

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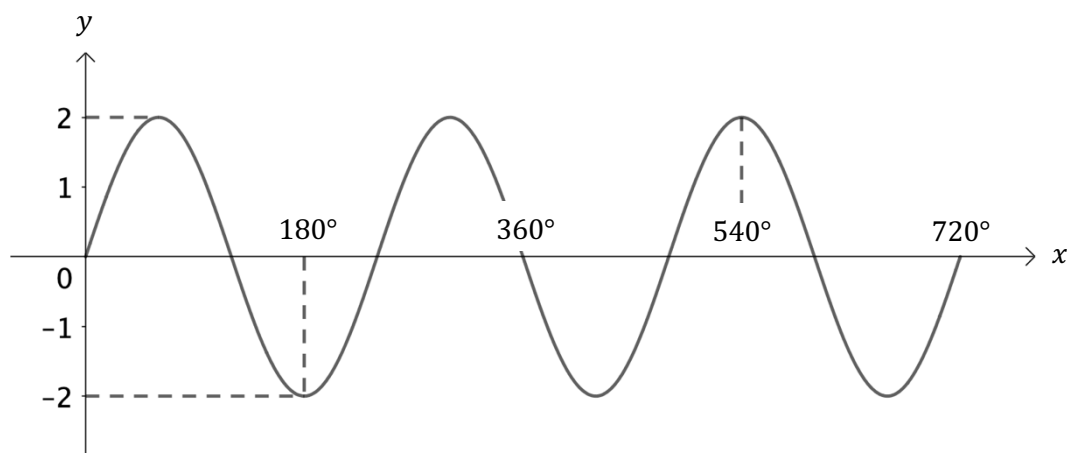
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(d) Find the equation of the trigonometric graph below.

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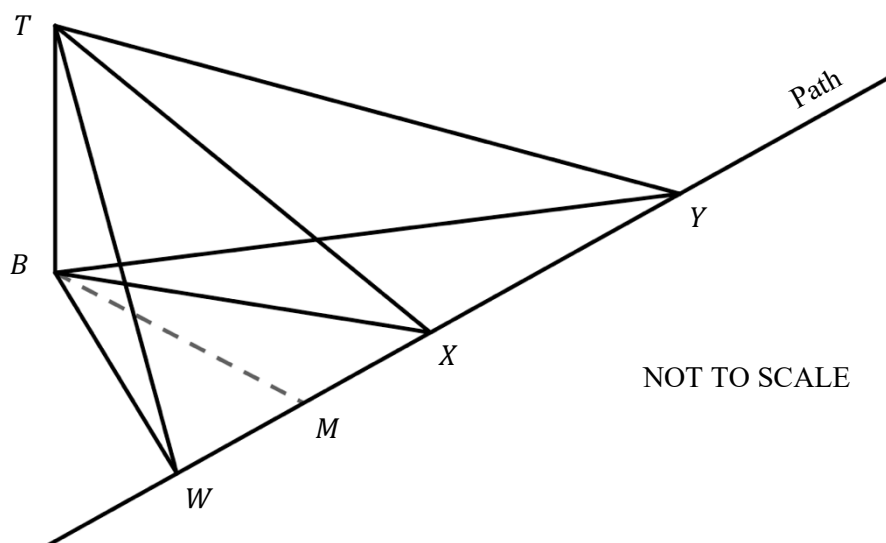
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Examination continues overleaf

- (e) Three points W, X and Y lie on a straight path on the ground, where $WX = XY = 40$ m. The base, B , of a flagpole, TB , is on the same horizontal plane as the path. M is the midpoint of WX . The angles of elevation to the top of the flagpole from the points W, X and Y are α, α , and β respectively. The height of the flagpole is h .



- i. Prove that $\triangle BWX$ is isosceles.

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- ii. Find the length of BM in terms of h and α .

1

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- iii. Hence, or otherwise, show that the height of the flagpole is given by

3

$$h = \frac{40\sqrt{2}}{\sqrt{\cot^2 \beta - \cot^2 \alpha}}$$

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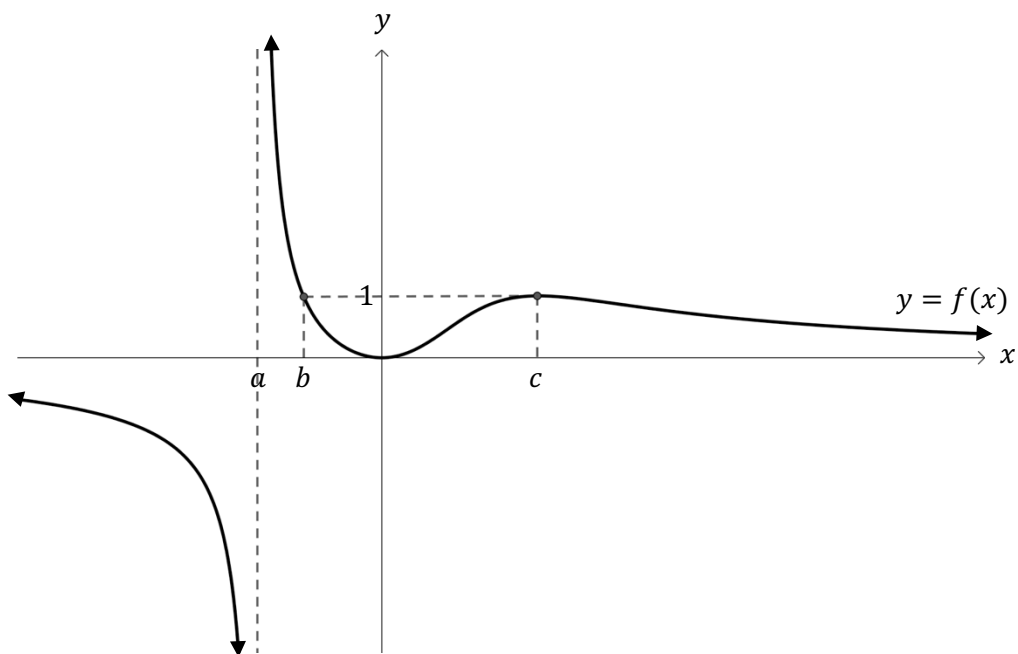
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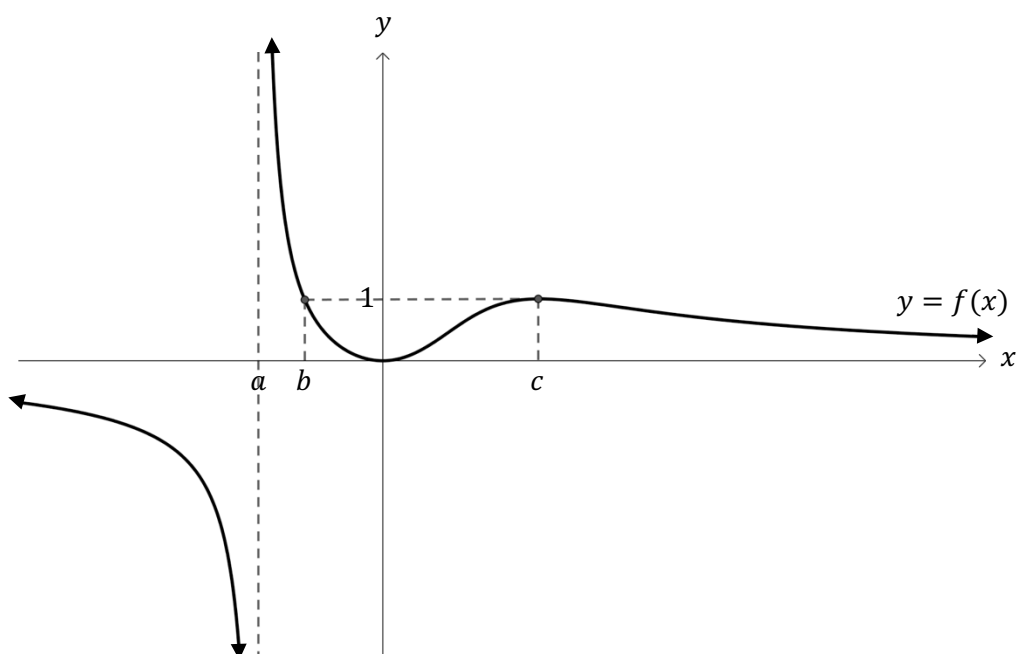
Examination continues overleaf

Question 4 (13 marks)**Marks**(a) Given the graph of $y = f(x)$, sketch the following on the same set of axes:

i. $y = f(-x)$

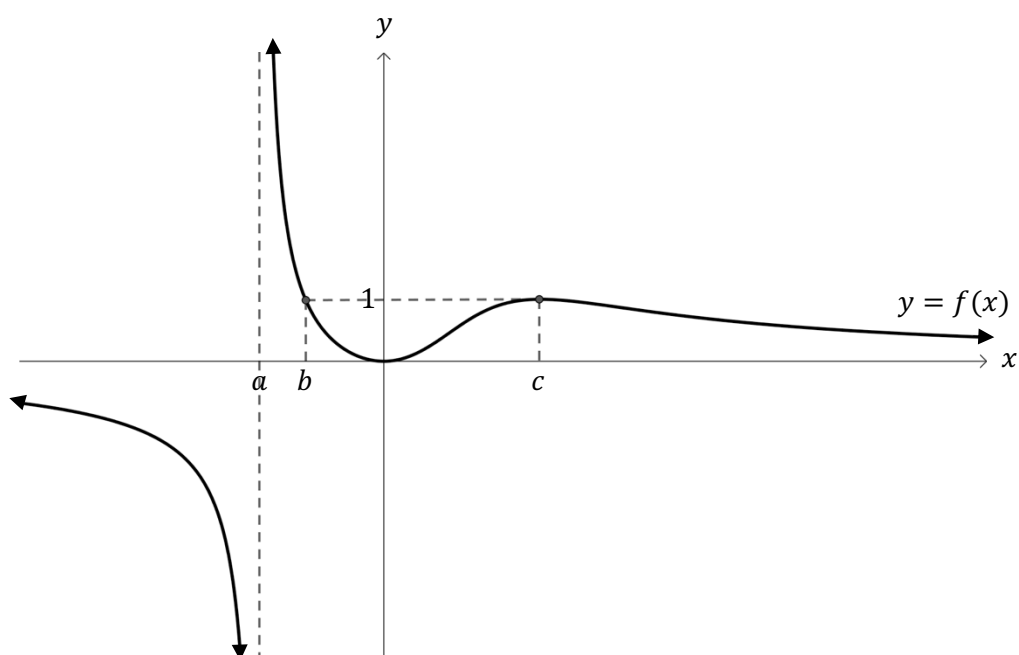
2

ii. $y = \frac{1}{f(x)}$

2

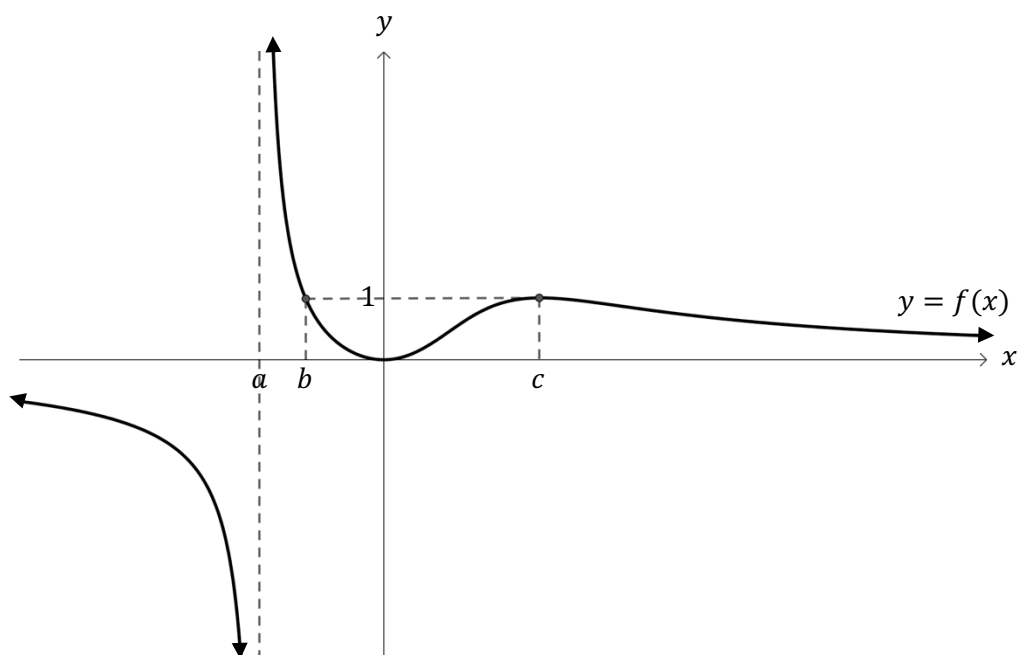
iii. $y^2 = f(x)$

2



iv. $y = f(|x|)$

2



(b) For the function $f(x) = x^2 - 4x + 4$.

- i. State the largest domain containing positive values for which $f(x)$ has an inverse. 1

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- ii. Find the equation of the inverse function. 2

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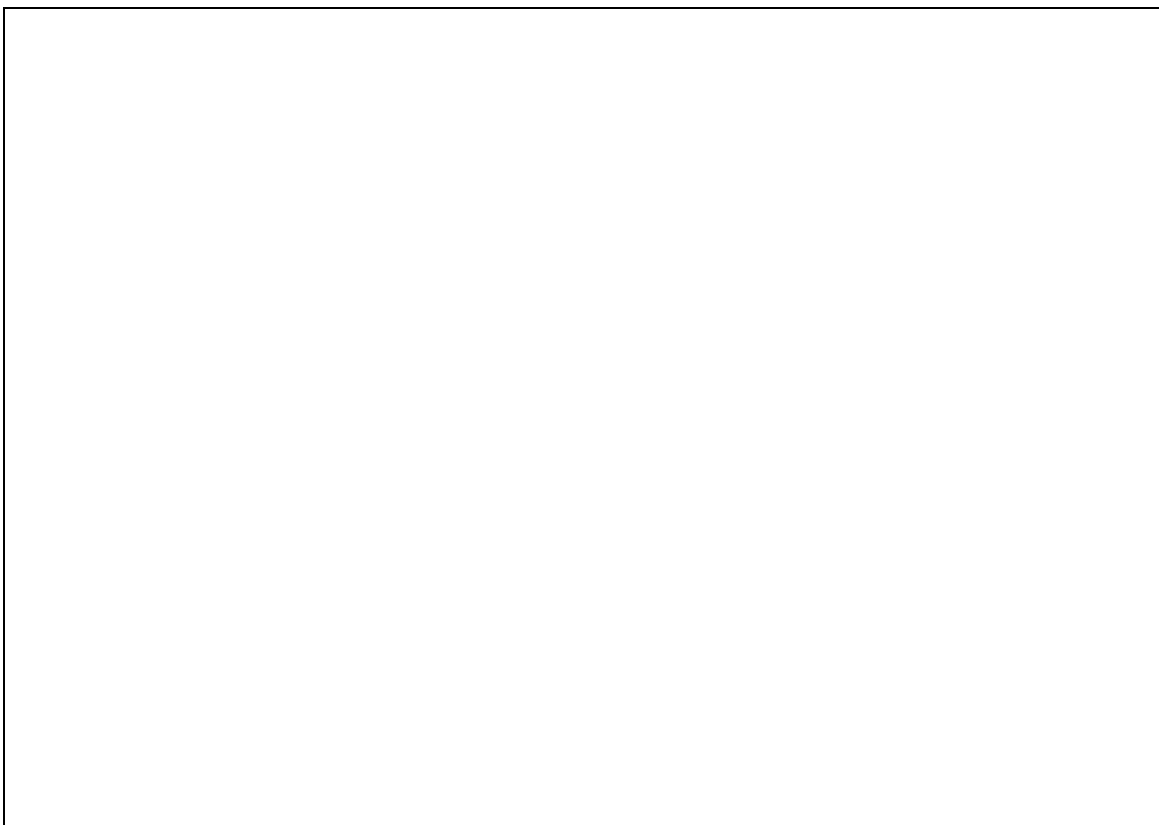
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- (c) i. Sketch the graph of $y = x$ and $y = 2^{-x}$ on the same axes. 1

- ii. Hence, sketch the graph of $y = x + 2^{-x}$ by addition of ordinates, showing all important features. 1



End of Examination



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Question 1 (22 marks)**Marks**

- (a) Is the function
- $f(x) = x^3 - x$
- odd, even or neither? Justify your answer.

2

$$\begin{aligned}
 f(-x) &= (-x)^3 - (-x) \\
 &= -x^3 + x \\
 &= -(x^3 - x) \\
 &= -f(x)
 \end{aligned}$$

$\therefore$ odd function ✓

- (b) Find the domain and range of the following in interval notation.

i. $y = 3x^2 - 1$

2

$$\begin{aligned}
 D: & (-\infty, \infty) \quad \checkmark \\
 R: & [-1, \infty) \quad \checkmark
 \end{aligned}$$

ii. $y = \frac{1}{x-2}$

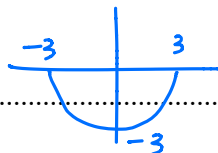
2

$$\begin{aligned}
 D: & (-\infty, 2) \cup (2, \infty) \quad \checkmark \\
 R: & (-\infty, 0) \cup (0, \infty) \quad \checkmark
 \end{aligned}$$

iii. $y = -\sqrt{9 - x^2}$

2

$$\begin{aligned}
 D: & [-3, 3] \quad \checkmark \\
 R: & [-3, 0] \quad \checkmark
 \end{aligned}$$



- (c) Given
- $f(x) = x^2 + 1$
- and
- $g(x) = 3 - x$
- , find:

i. $f(g(2))$.

1

$$\begin{aligned}
 g(2) &= 3 - 2 = 1 \\
 f(g(2)) &= 1^2 + 1 = 2 \quad \checkmark
 \end{aligned}$$

ii. $g(f(x))$ in simplest form.

1

$$\begin{aligned}
 g(f(x)) &= 3 - (x^2 + 1) \\
 &= 2 - x^2 \quad \checkmark
 \end{aligned}$$

deduct 1 mark overall if not in interval notation

- iii. The value(s) of x such that $g(f(x)) = f(x)$.

2

$$2 - x^2 = x^2 + 1 \quad \checkmark$$

$$0 = 2x^2 - 1$$

$$x^2 = \frac{1}{2}$$

$$x = \pm \frac{1}{\sqrt{2}} \quad \checkmark$$

- (d) Solve $|7 - 3x| = 1$

2

$$7 - 3x = 1 \quad , \quad 7 - 3x = -1$$

$$6 = 3x$$

$$8 = 3x$$

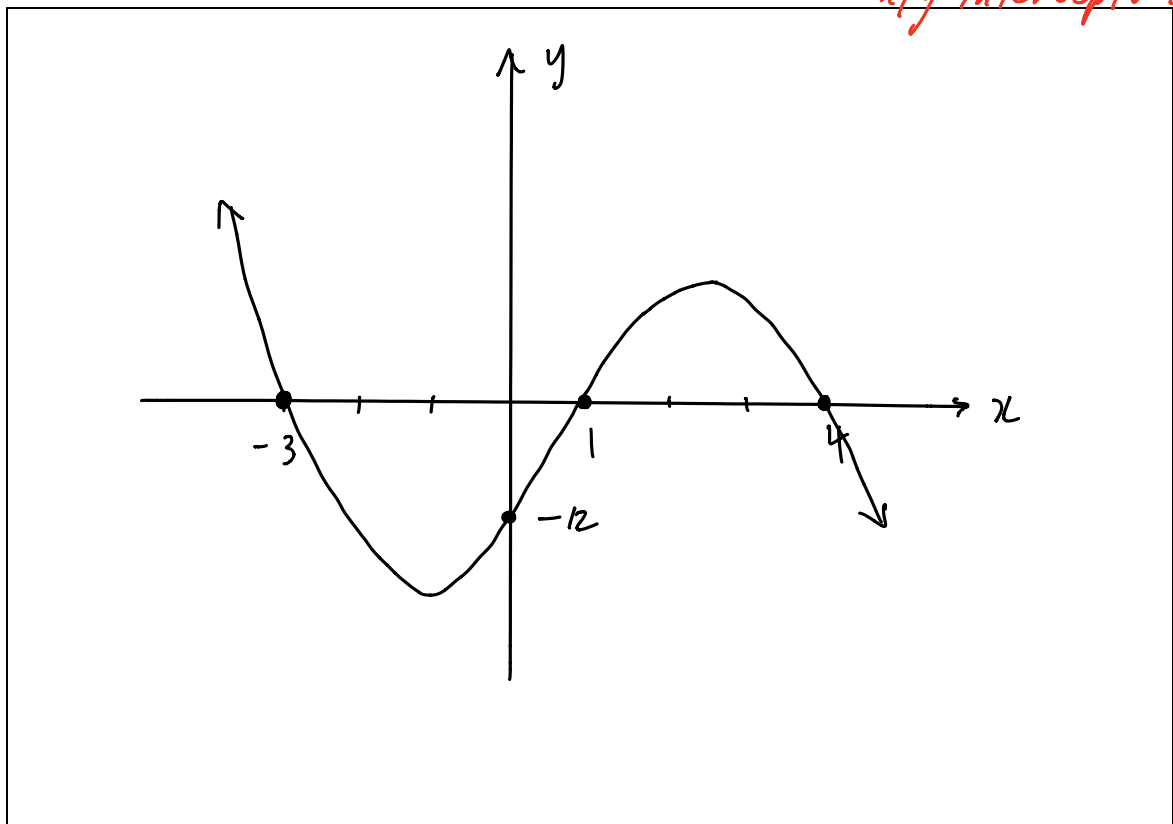
$$x = 2 \quad \checkmark$$

$$x = \frac{8}{3} \quad \checkmark$$

- (e) Sketch the following graphs showing all important features, including x and y intercepts.

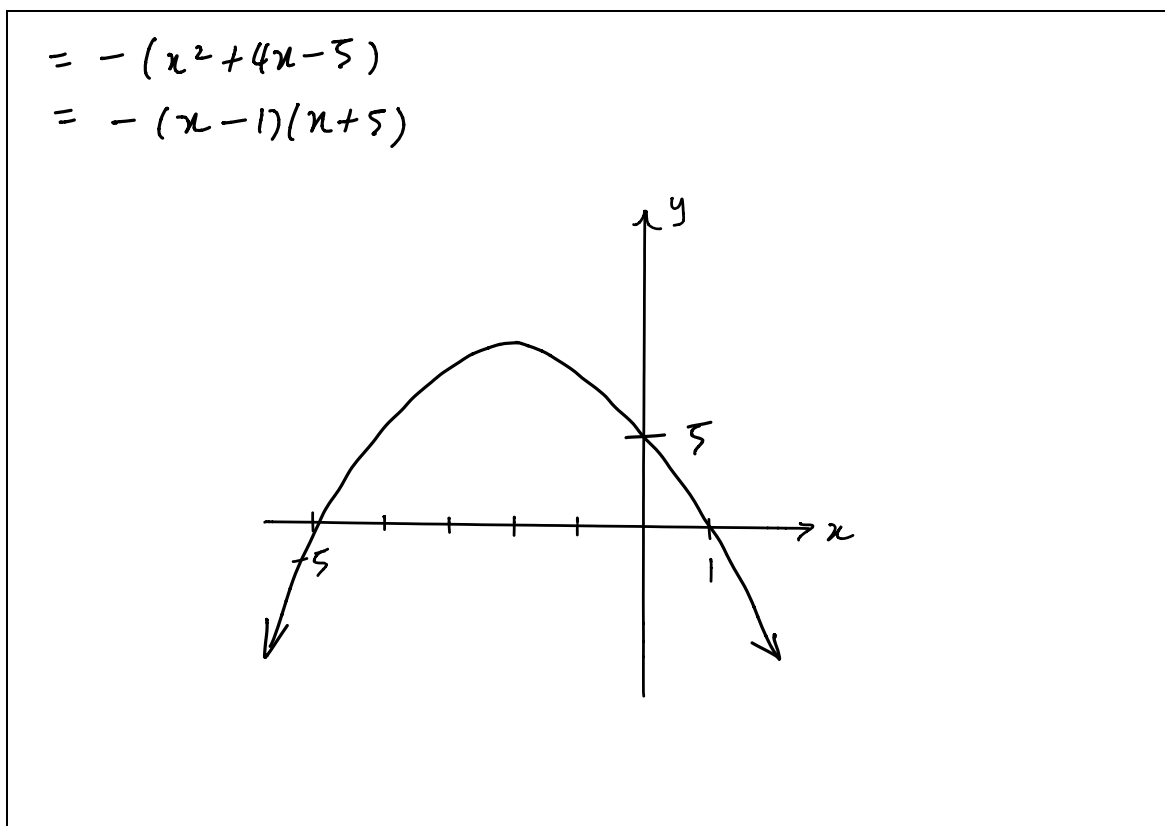
i. $y = (x - 1)(x + 3)(4 - x)$

*✓ shape ✓ critical point+ /
x/y intercepts labelled*



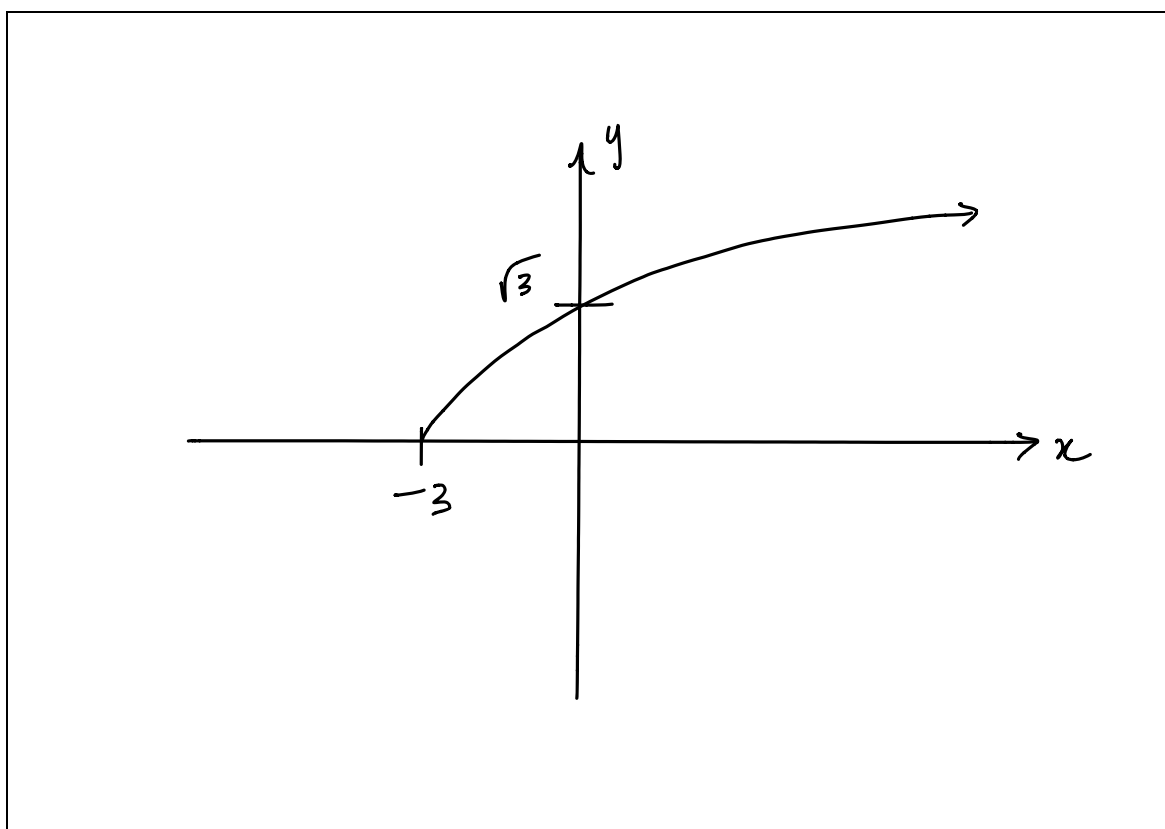
ii. $y = -x^2 - 4x + 5$

2



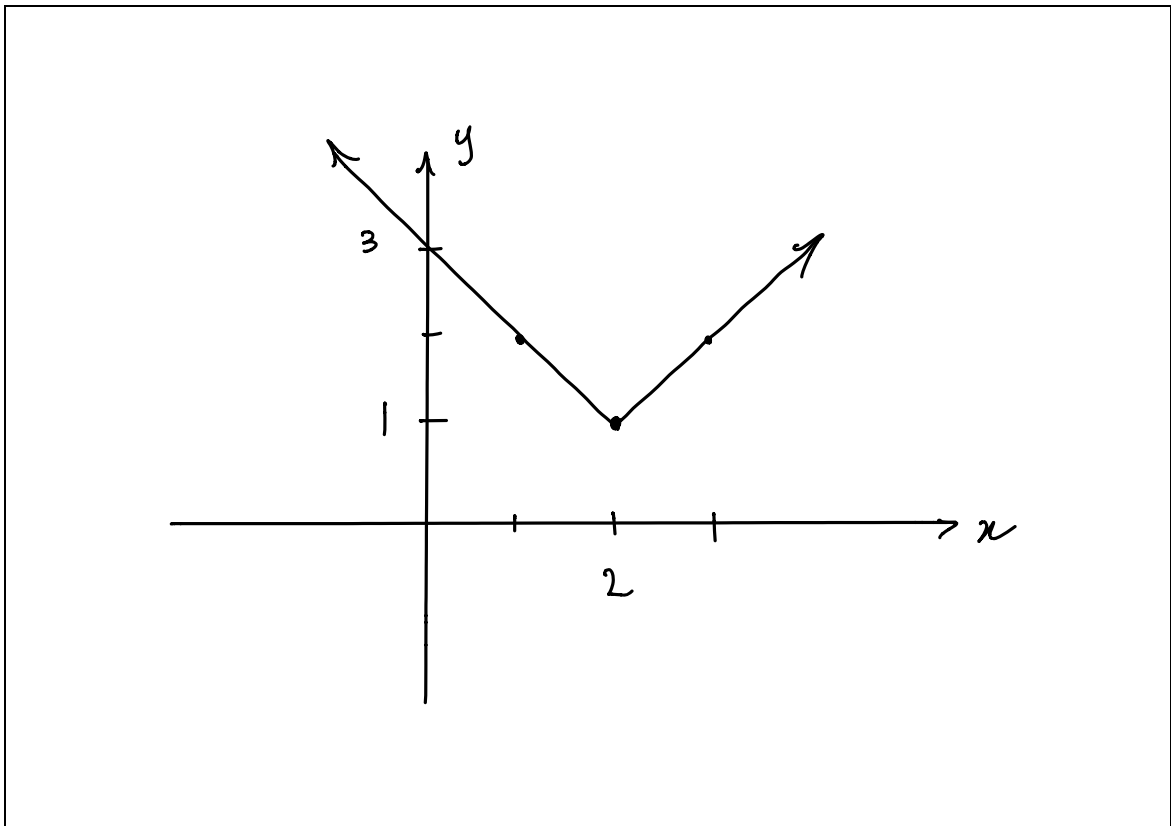
iii. $y = \sqrt{x + 3}$

2



iv. $y = |x - 2| + 1$

2

**Examination continues overleaf**

Question 2 (11 marks)**Marks**

- (a) Solve the equation
- $\log_2 x + \log_2(x+7) = 3$
- .

3

$$\log_2 (x(x+7)) = 3 \quad \checkmark$$

$$2^3 = x^2 + 7x$$

$$0 = x^2 + 7x - 8$$

$$= (x-1)(x+8)$$

$$x = 1, -8 \quad \checkmark$$

$$= 1 \quad \checkmark \quad (x+7 > 0 \therefore x > -7)$$

- (b) Fully simplify
- $\sqrt{\frac{3^{5k+2}}{27^k}}$

2

$$= \left(\frac{3^{5k+2}}{3^{3k}} \right)^{\frac{1}{2}} \quad \checkmark$$

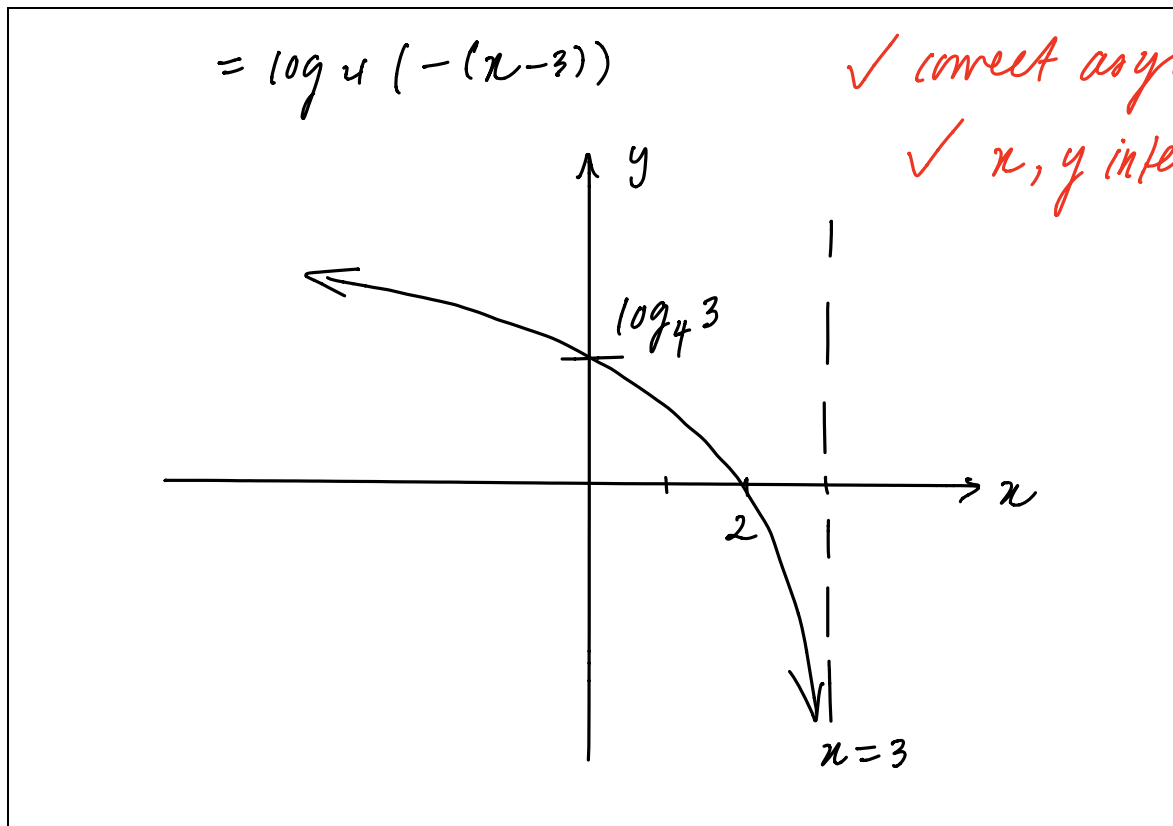
$$= (3^{5k+2-3k})^{\frac{1}{2}}$$

$$= (3^{2k+2})^{\frac{1}{2}}$$

$$= 3^{k+1} \quad \checkmark$$

- (c) Sketch the graph of $y = \log_4(3 - x)$, showing all important features.

2



- (d) The number N of individuals in a population at time t years is given by

$$N = 100 - 60e^{-0.1t} \quad \text{where } e = 2.71828 \dots$$

- i. Find the initial population N at $t = 0$ and what N tends towards as $t \rightarrow \infty$.

2

$$\begin{aligned} N &= 100 - 60e^0 \\ &= 40 \quad \checkmark \\ \text{as } t \rightarrow \infty, N &\rightarrow 100 \quad \checkmark \quad (\text{since } e^{-0.1t} \rightarrow 0) \end{aligned}$$

- ii. Find the exact time taken for the population to double its initial size.

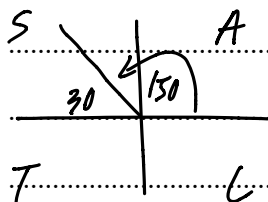
2

$$\begin{aligned} 80 &= 100 - 60e^{-0.1t} \\ -20 &= -60e^{-0.1t} \\ \frac{1}{3} &= e^{-0.1t} \quad \checkmark \\ \log_e \frac{1}{3} &= -0.1t \\ t &= \frac{1}{-0.1} \log_e \frac{1}{3} \quad \checkmark \\ &= 10 \log_e 3 \end{aligned}$$

Question 3 (14 marks)**Marks**

- (a) If
- $\sin \theta = \frac{1}{2}$
- and
- $\cos \theta < 0$
- , what is the exact value of
- $\tan \theta$
- ?

2



$$\tan \theta = -\frac{1}{\sqrt{3}}$$

- (b) Solve
- $\cos^2 x + 2\sin x = 0$
- , for
- $0^\circ \leq x \leq 360^\circ$
- .

2

$$1 - \sin^2 x + 2\sin x = 0$$

$$\sin^2 x - 2\sin x - 1 = 0$$

$$u^2 - 2u - 1 = 0$$

$$u = 1 \pm \sqrt{2}$$

$$\sin x = 1 \pm \sqrt{2}$$

$$= 1 - \sqrt{2} \text{ only since } -1 \leq \sin x \leq 1$$

$$= -0.414\dots$$

$$= 204^\circ 28', 335^\circ 32'$$

correct to the nearest
minute

accept correct to five
nearest degree also.

- (c) Solve
- $\tan 2x = -\sqrt{3}$
- , for
- $-180^\circ \leq x \leq 180^\circ$
- .

2

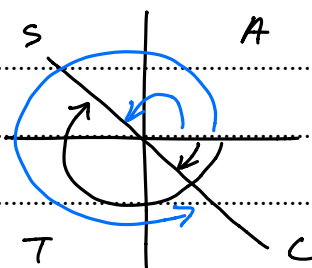
$$-360^\circ \leq 2x \leq 360^\circ$$

$$2x = \tan^{-1}(-\sqrt{3})$$

$$= -60^\circ, -240^\circ,$$

$$120^\circ, 300^\circ$$

$$x = -30^\circ, -120^\circ, 60^\circ, 150^\circ$$

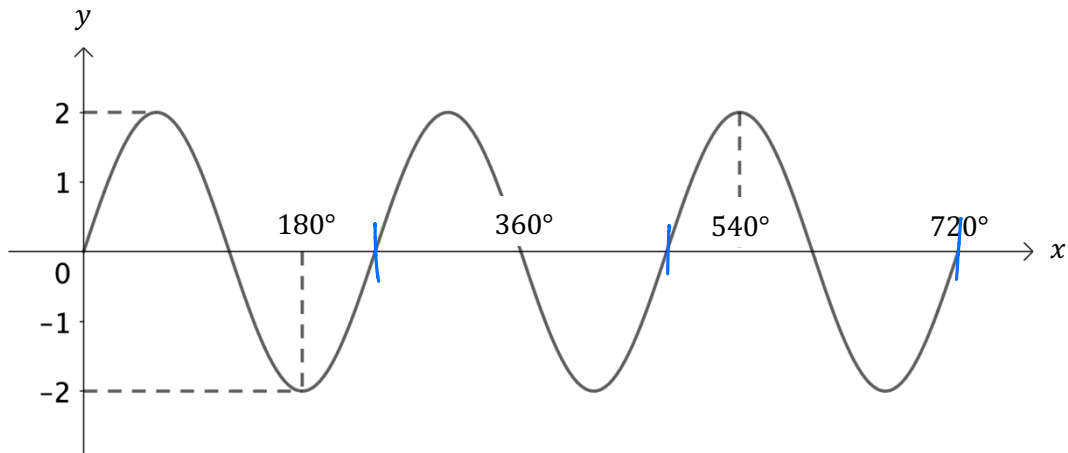


✓ for $-30^\circ, 60^\circ$

✓ for $-120^\circ, 150^\circ$

(d) Find the equation of the trigonometric graph below.

2



$$y = 2 \sin(n\pi), \quad (180^\circ, -2) \rightarrow -2 = 2 \sin(n \times 180^\circ)$$

$$-1 = \sin(n \times 180^\circ)$$

$$n \times 180^\circ = \sin^{-1}(-1)$$

Alternatively,

$$3 \times \text{period} = 720^\circ$$

$$\text{period} = 240^\circ$$

$$\frac{2 \times 180^\circ}{n} = 240^\circ$$

$$\frac{3}{2} = n$$

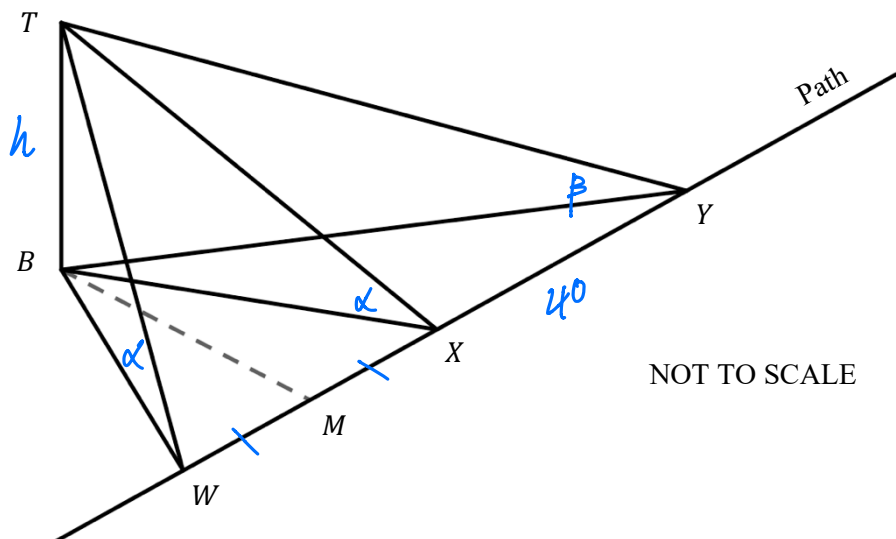
$$= 270^\circ$$

$$n = \frac{3}{2}$$

$$\therefore y = 2 \sin\left(\frac{3}{2}x\right)$$

Examination continues overleaf

- (e) Three points W, X and Y lie on a straight path on the ground, where $WX = XY = 40$ m. The base, B , of a flagpole, TB , is on the same horizontal plane as the path. M is the midpoint of WX . The angles of elevation to the top of the flagpole from the points W, X and Y are α, α , and β respectively. The height of the flagpole is h .



- i. Prove that $\triangle BWX$ is isosceles.

2

In $\triangle TBW$, $\tan \alpha = \frac{h}{BW} \rightarrow BW = h \cot \alpha$

In $\triangle TBX$, $\tan \alpha = \frac{h}{BX} \rightarrow BX = h \cot \alpha$

$\therefore BW = BX$

$\therefore \triangle BWX$ is isosceles

- ii. Find the length of BM in terms of h and α .

1

In $\triangle TBW$, $\tan \alpha = \frac{h}{BW}$

$BW = h \cot \alpha$

In $\triangle BWM$, $BM = \sqrt{BW^2 - 20^2}$

$= \sqrt{h^2 \cot^2 \alpha - 400}$

or $= \sqrt{\frac{h^2}{\tan^2 \alpha} - 400}$

iii. Hence, or otherwise, show that the height of the flagpole is given by

3

$$h = \frac{40\sqrt{2}}{\sqrt{\cot^2 \beta - \cot^2 \alpha}}$$

In $\triangle TBY$, $\tan \beta = \frac{h}{BY}$

$$BY = h \cot \beta \quad \checkmark$$

In $\triangle BMY$, $\angle BMY = 90^\circ$ ($\triangle BWX$ isosceles)

$$\therefore BY^2 = BM^2 + MY^2 \quad (\text{Pythagoras' theorem})$$

$$h^2 \cot^2 \beta = h^2 \cot^2 \alpha - 400 + 60^2 \quad \checkmark$$

$$h^2 (\cot^2 \beta - \cot^2 \alpha) = 3200$$

$$h = \sqrt{\frac{3200}{\cot^2 \beta - \cot^2 \alpha}}$$

$$= \frac{40\sqrt{2}}{\sqrt{\cot^2 \beta - \cot^2 \alpha}}$$

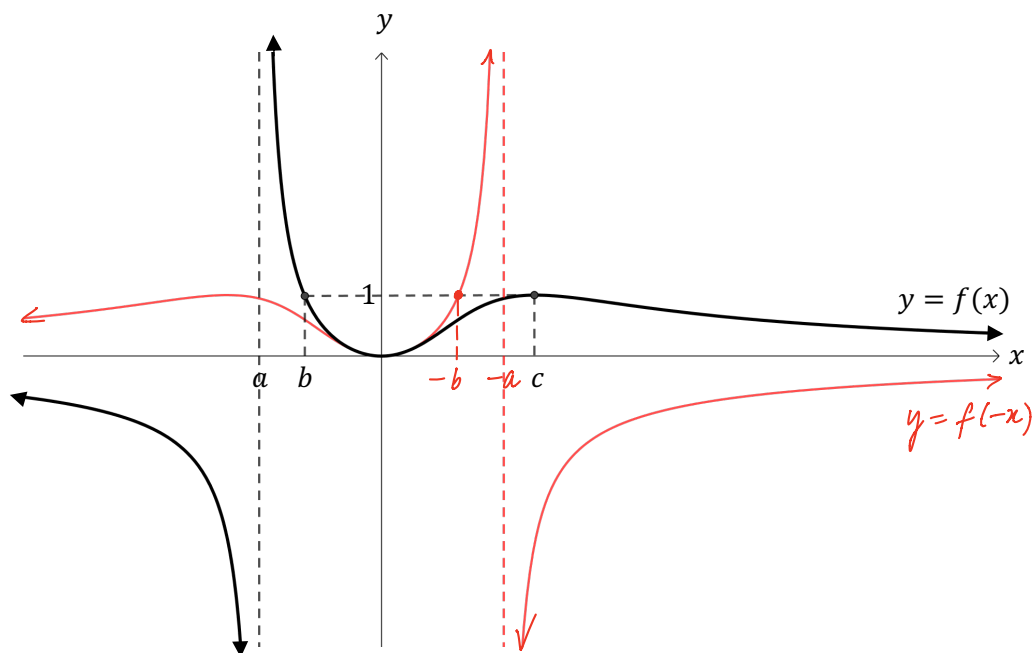
$(h > 0)$

Examination continues overleaf

Question 4 (13 marks)**Marks**(a) Given the graph of $y = f(x)$, sketch the following on the same set of axes:

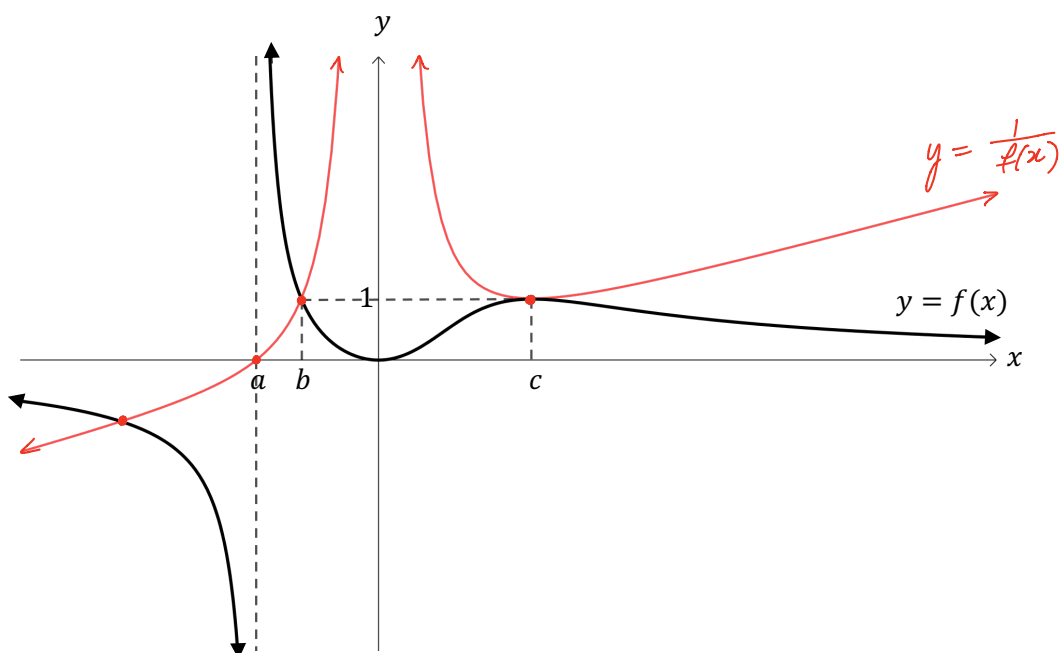
i. $y = f(-x)$

2



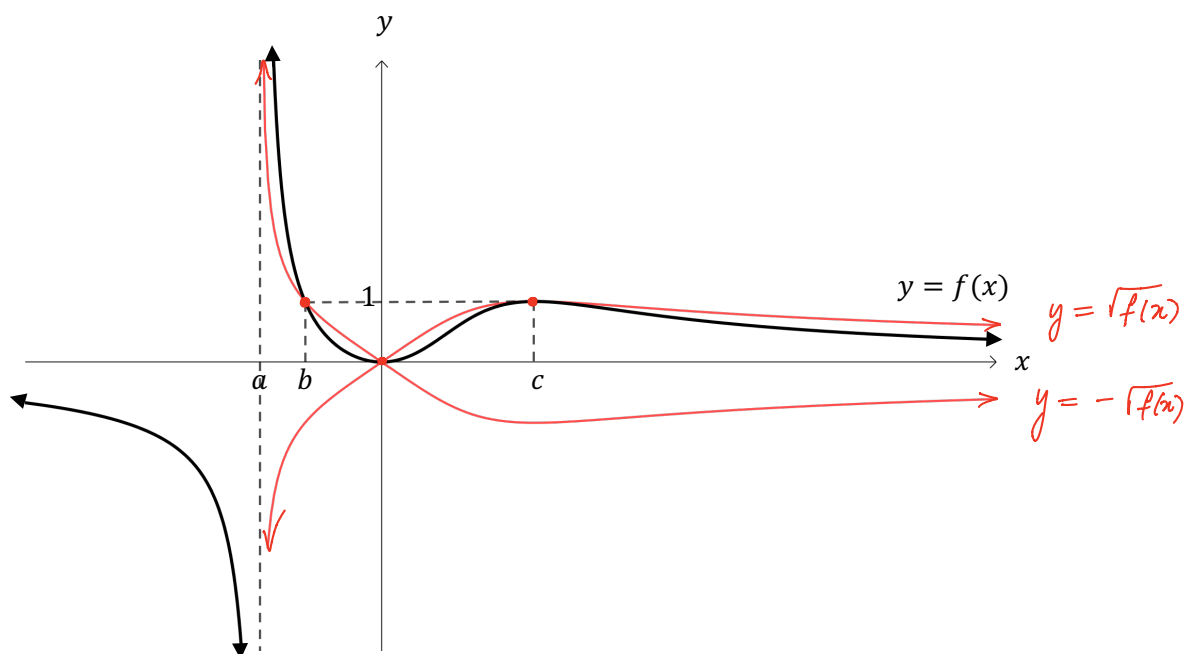
ii. $y = \frac{1}{f(x)}$

2



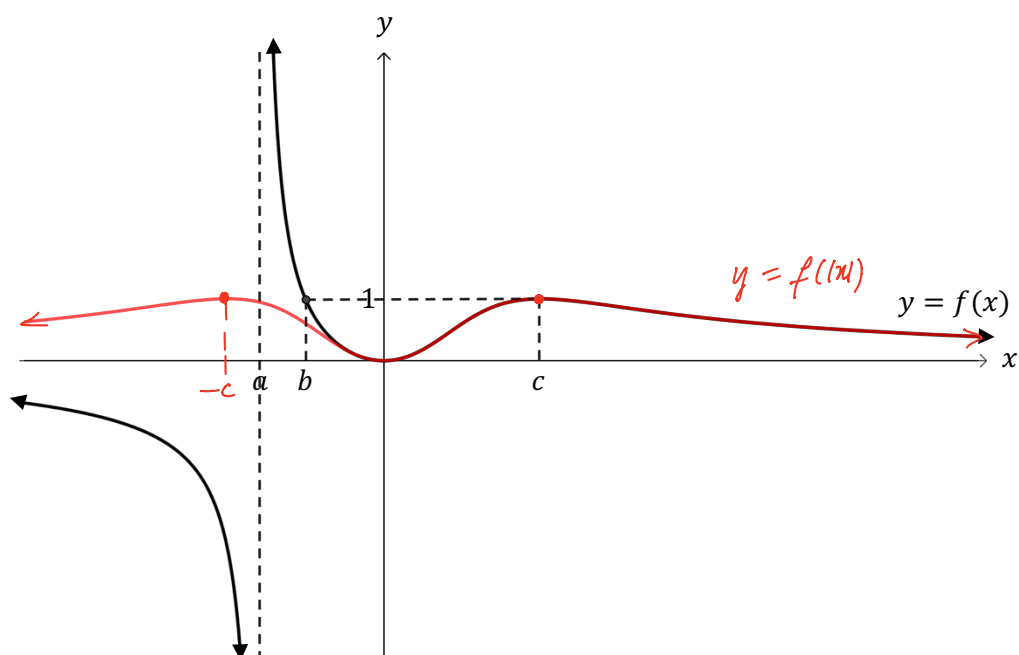
iii. $y^2 = f(x)$

2



iv. $y = f(|x|)$

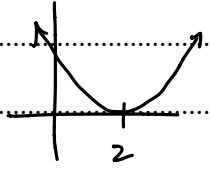
2



(b) For the function $f(x) = x^2 - 4x + 4$.

i. State the largest domain containing positive values for which $f(x)$ has an inverse. 1

$f(x) = (x-2)^2$
 $D: x \in \mathbb{R}, x \geq 2$ ✓

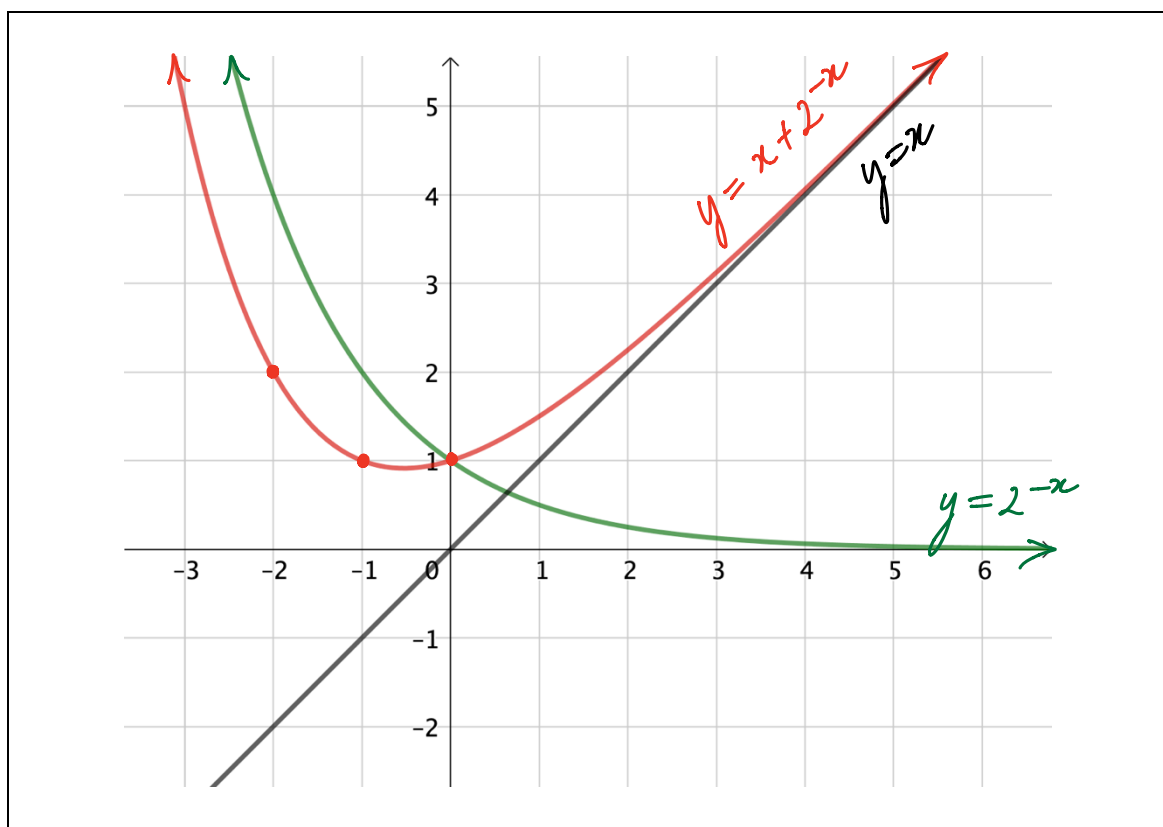


ii. Find the equation of the inverse function. 2

$x = (y-2)^2$
 $\pm \sqrt{x} = y-2$
 $y = \pm \sqrt{x} + 2$ ✓
 $= \sqrt{x} + 2$ ✓ (since $y \geq 2$) since the domain of $f(x)$ becomes range of $f^{-1}(x)$.

(c) i. Sketch the graph of $y = x$ and $y = 2^{-x}$ on the same axes. 1

ii. Hence, sketch the graph of $y = x + 2^{-x}$ by addition of ordinates, showing all important features. 1



End of Examination